

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

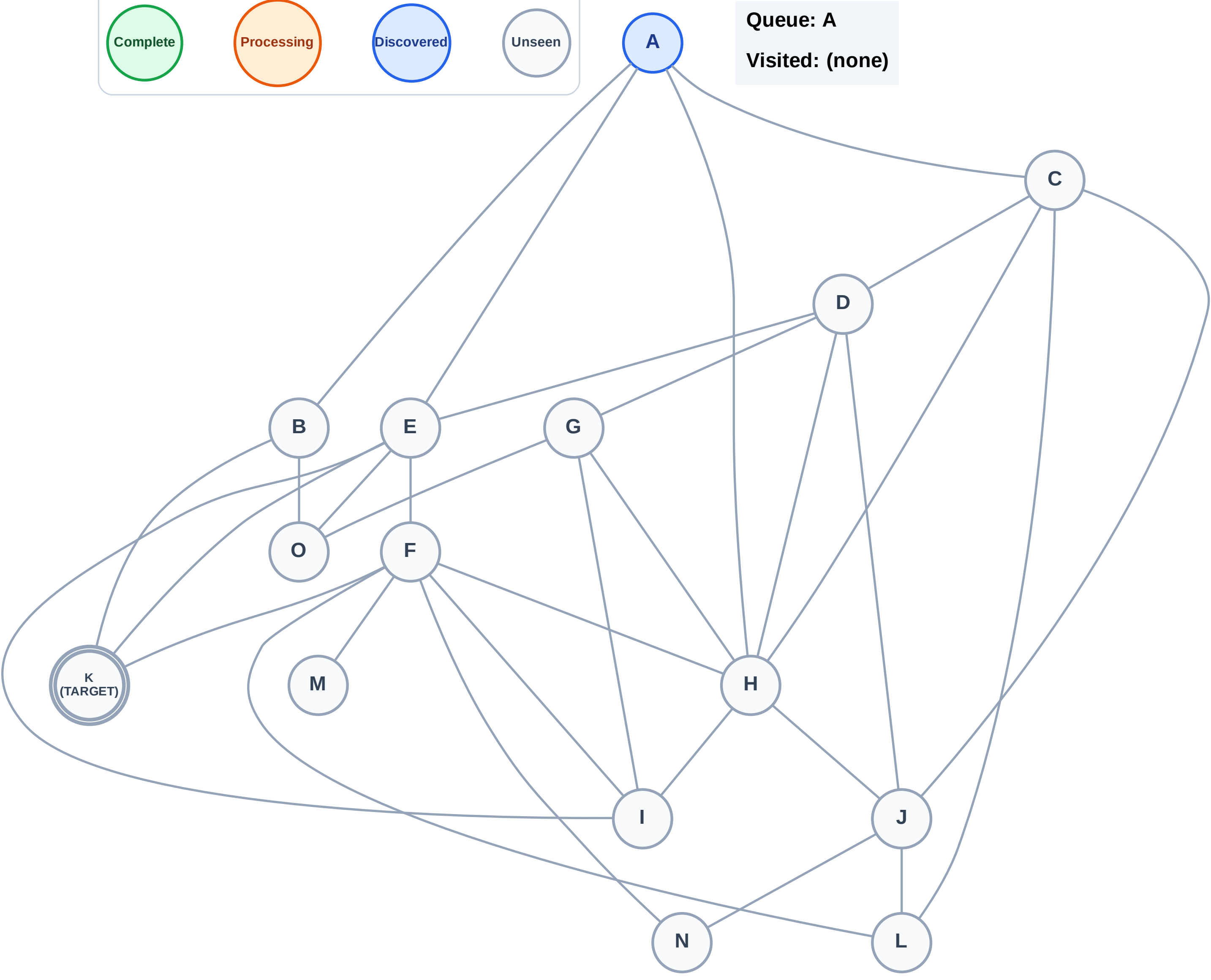
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

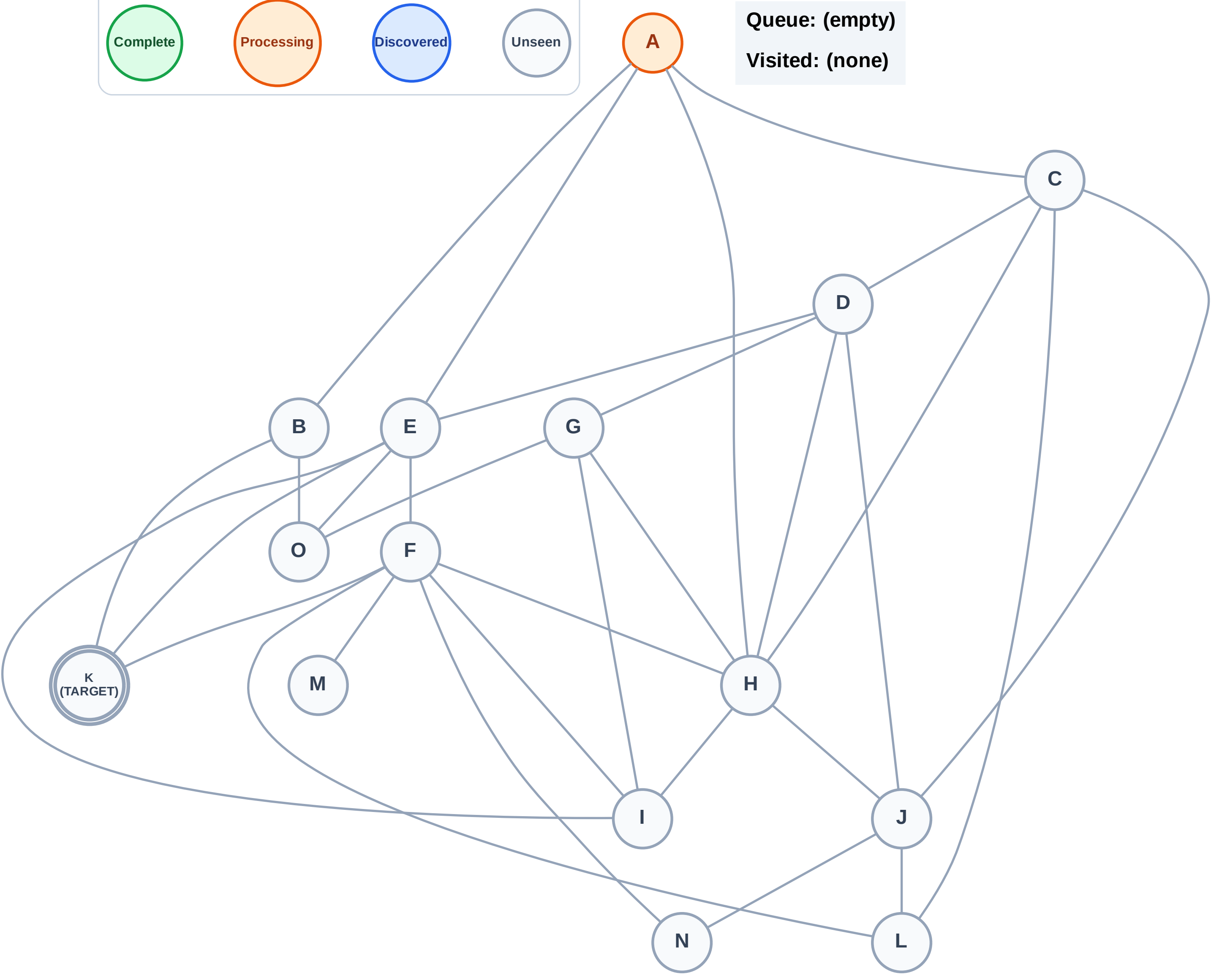
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, H from A

Legend

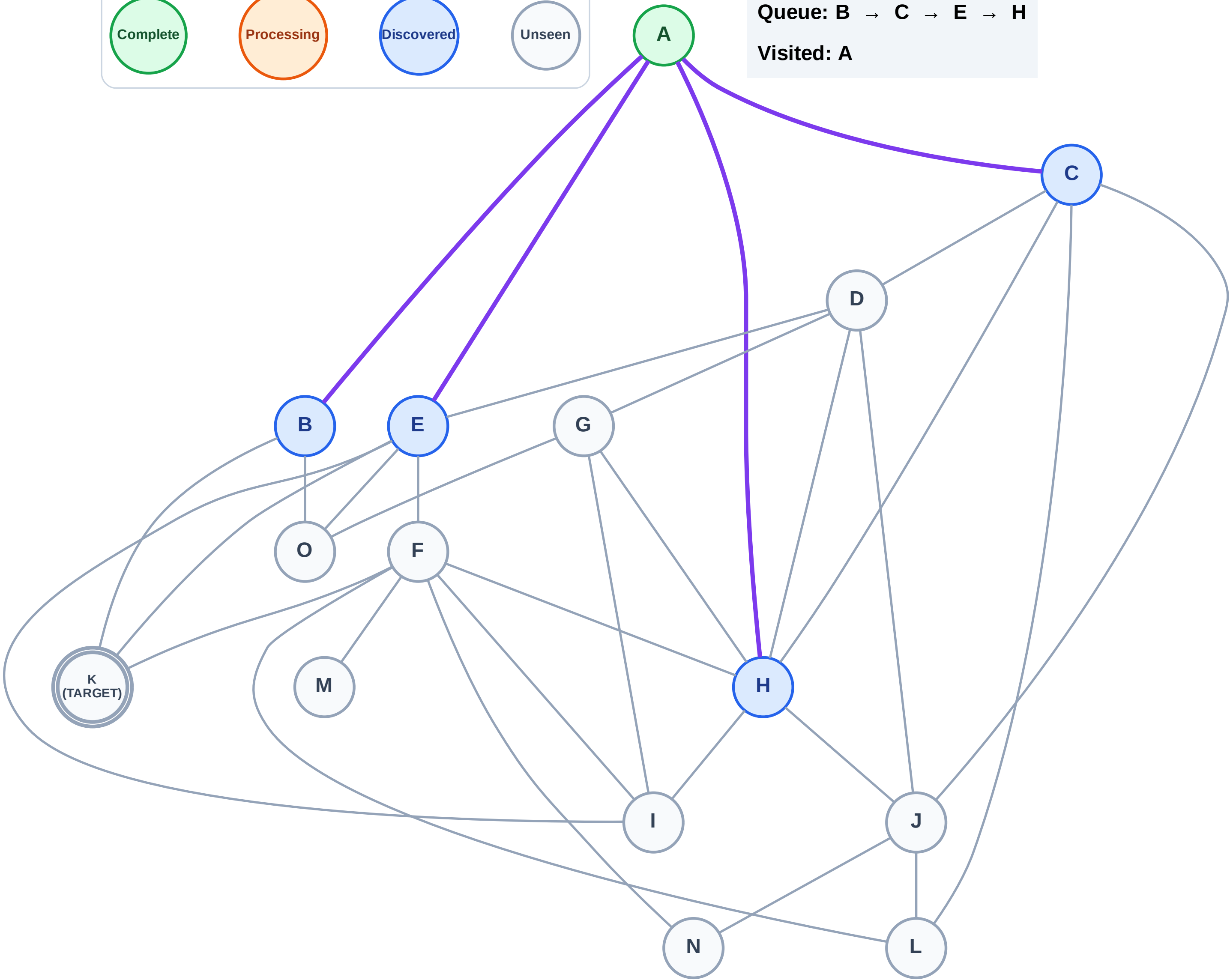
Complete

Processing

Discovered

Unseen

Queue: B → C → E → H
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

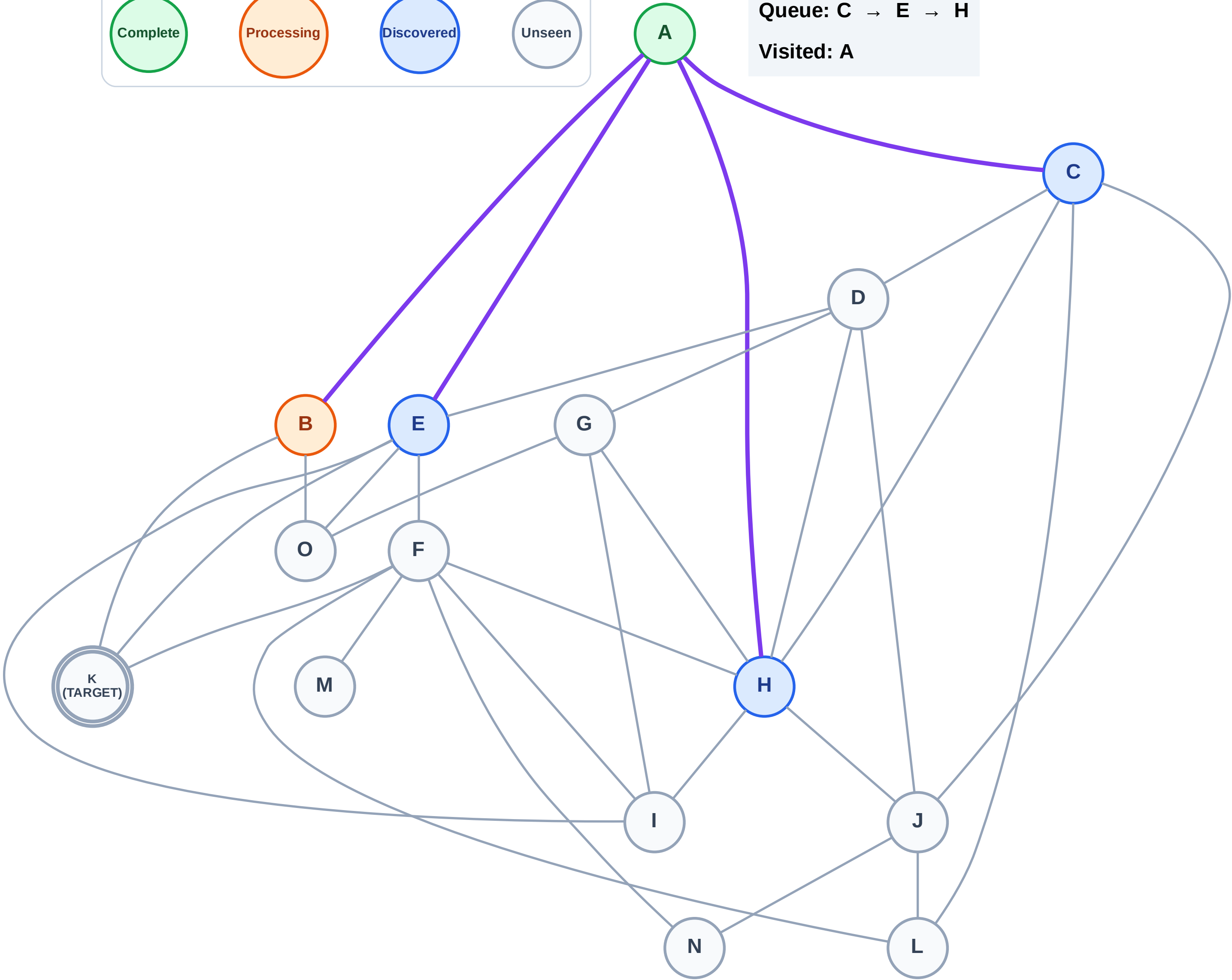
Processing

Discovered

Unseen

Queue: C → E → H

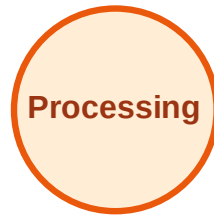
Visited: A



BFS Traversal

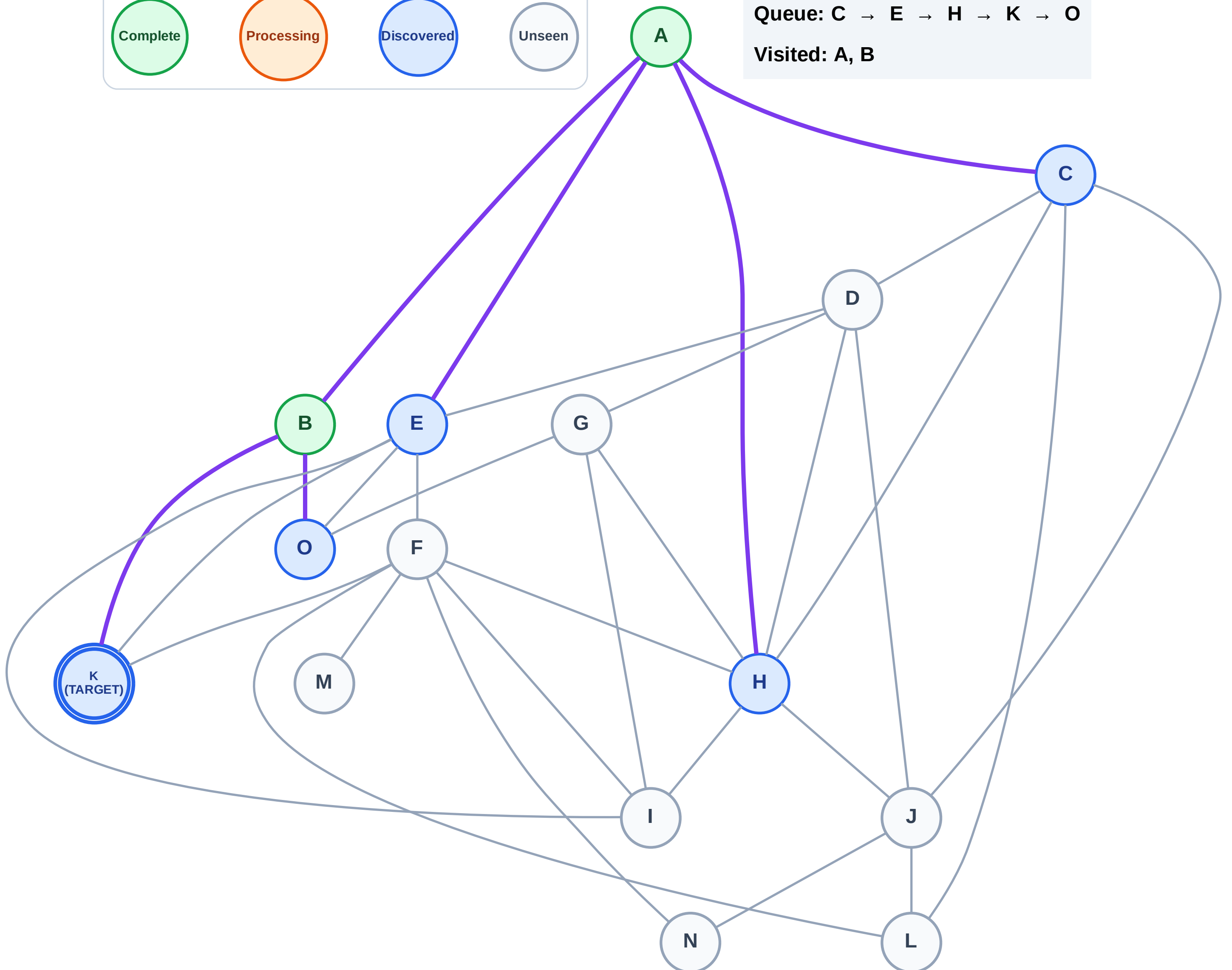
Step 5: Discover K, O from B

Legend



Queue: C → E → H → K → O

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

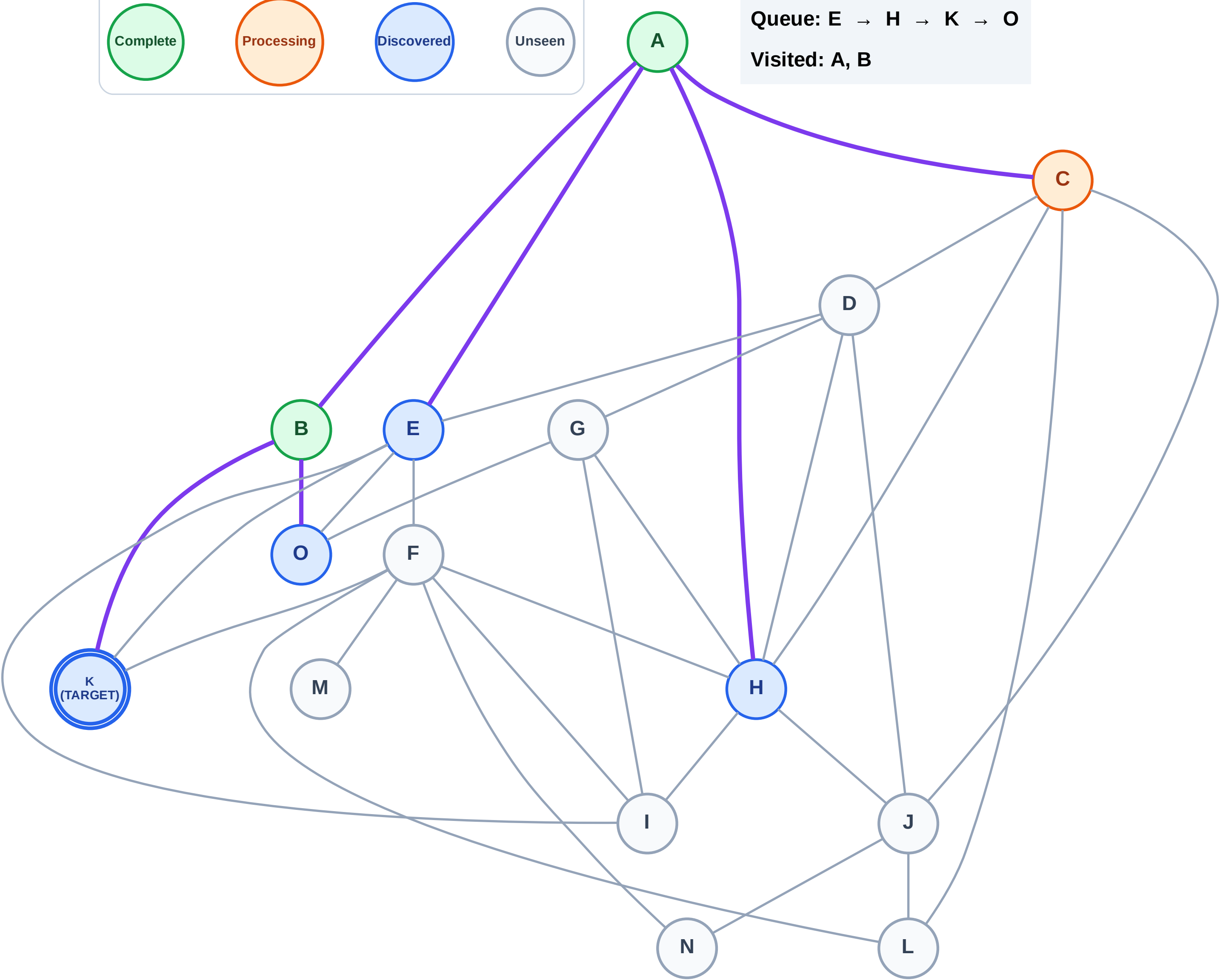
Complete

Processing

Discovered

Unseen

Queue: E → H → K → O
Visited: A, B



BFS Traversal

Step 7: Discover D, J, L from C

Legend

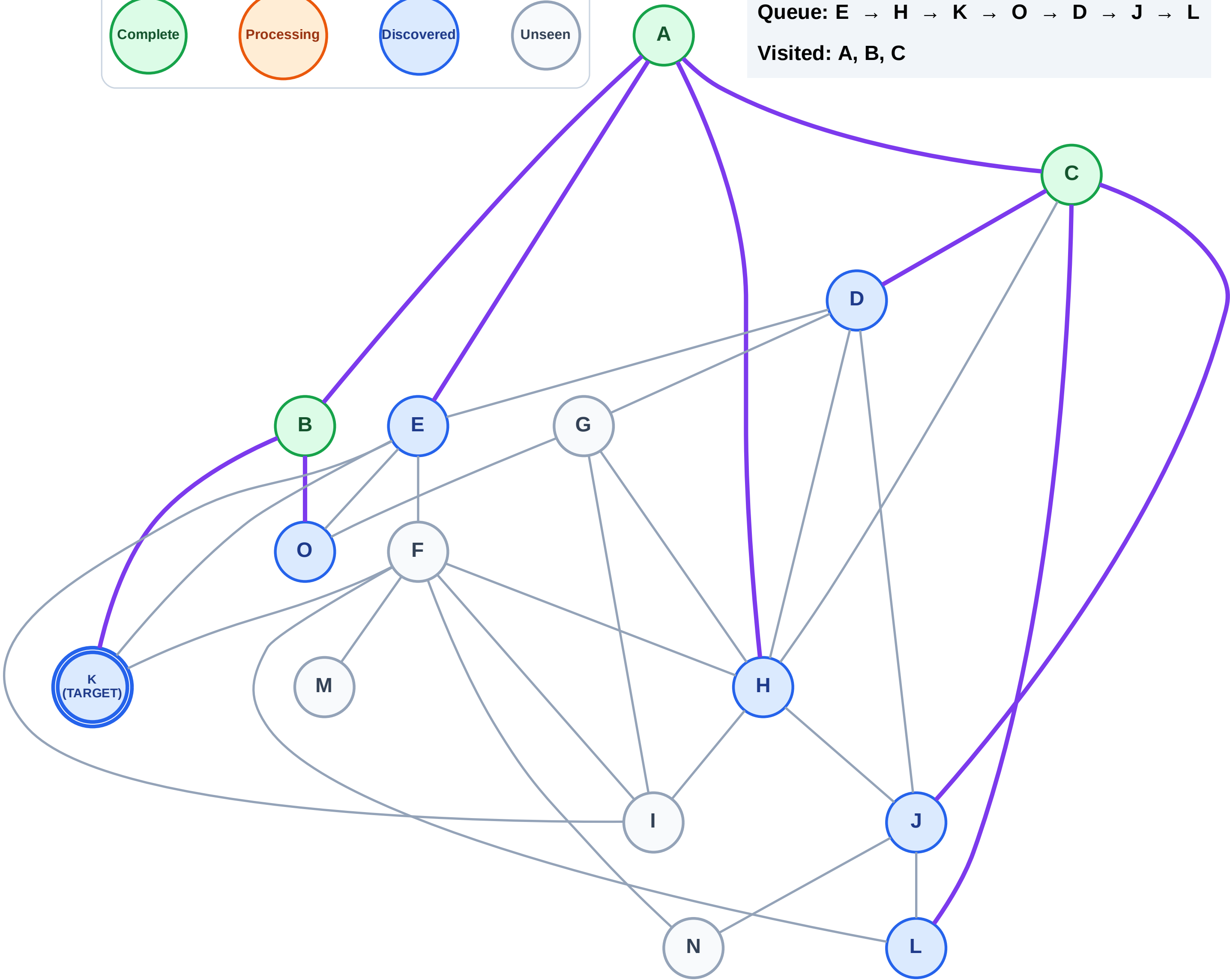
Complete

Processing

Discovered

Unseen

Queue: E → H → K → O → D → J → L
Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

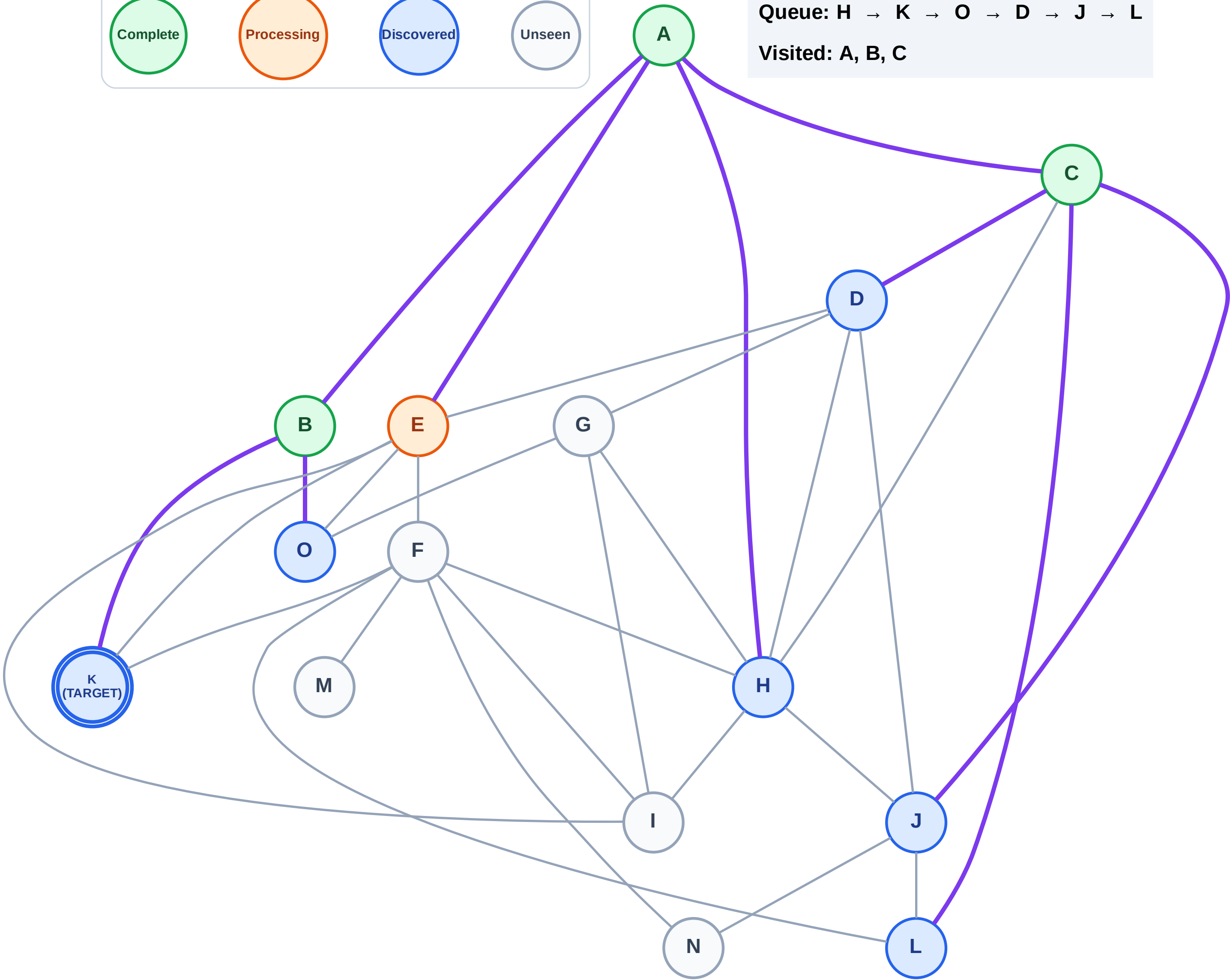
Processing

Discovered

Unseen

Queue: H → K → O → D → J → L

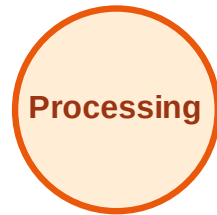
Visited: A, B, C



BFS Traversal

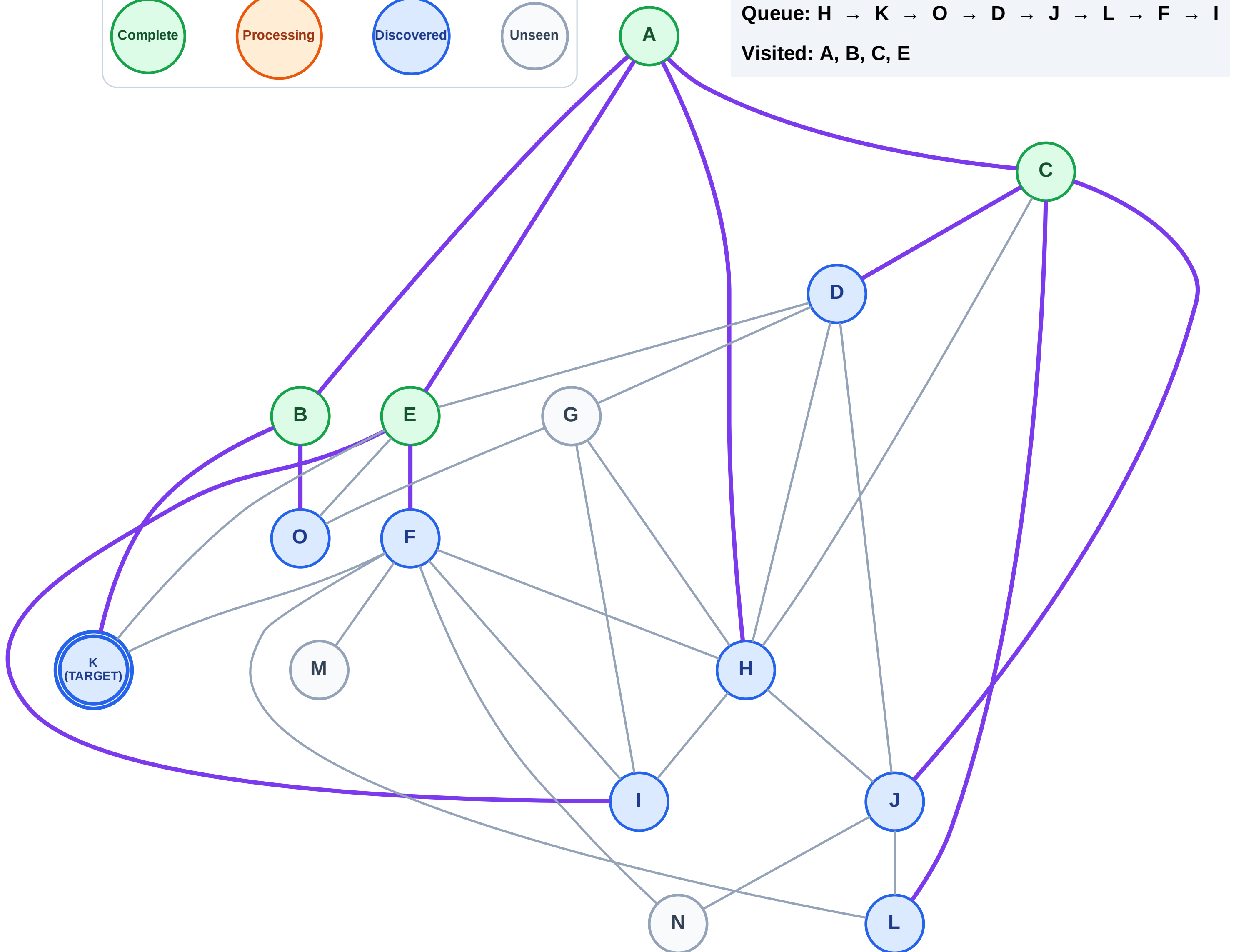
Step 9: Discover F, I from E

Legend



Queue: H → K → O → D → J → L → F → I

Visited: A, B, C, E



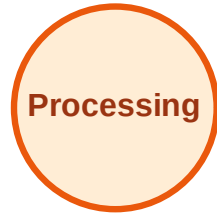
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



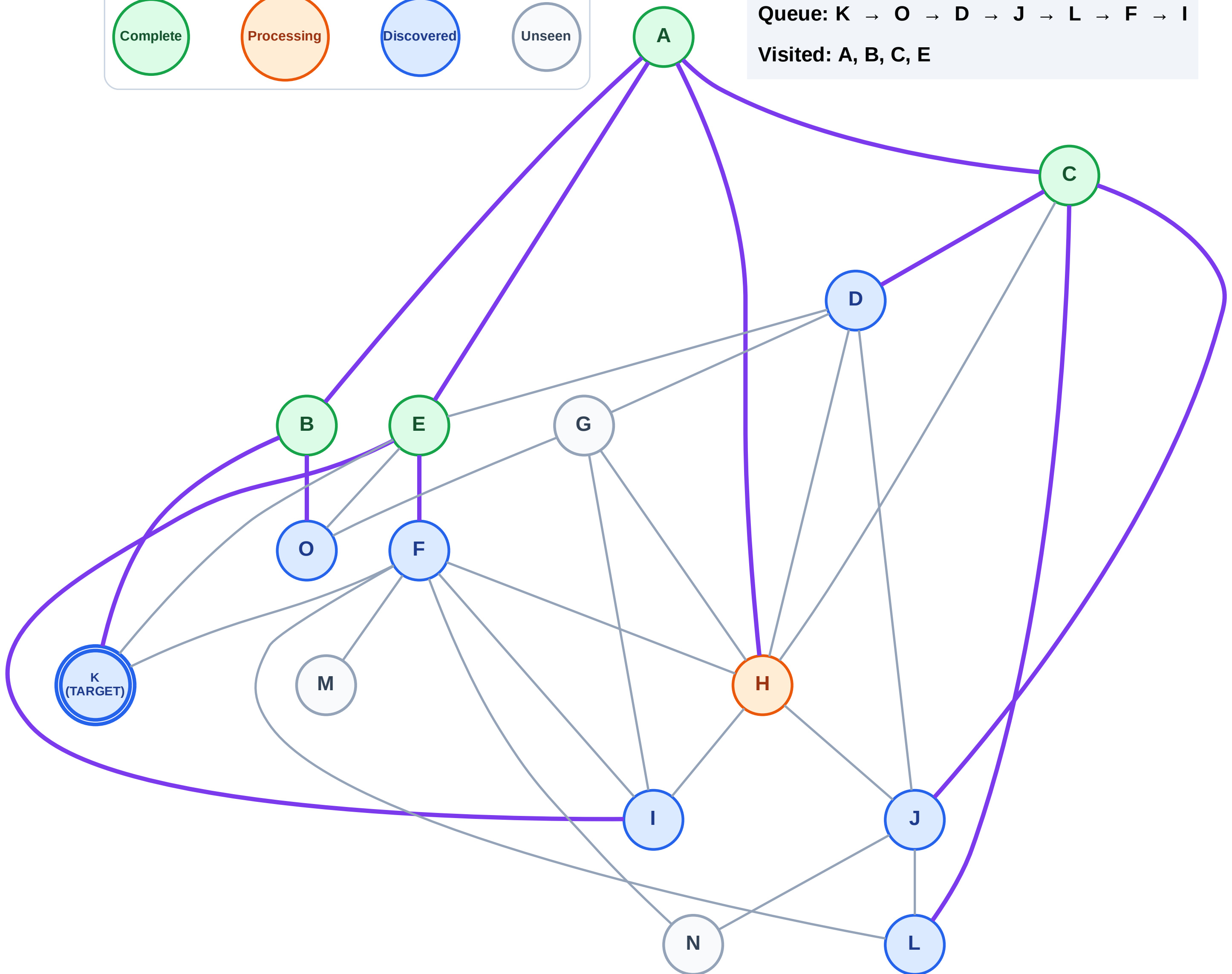
Discovered



Unseen

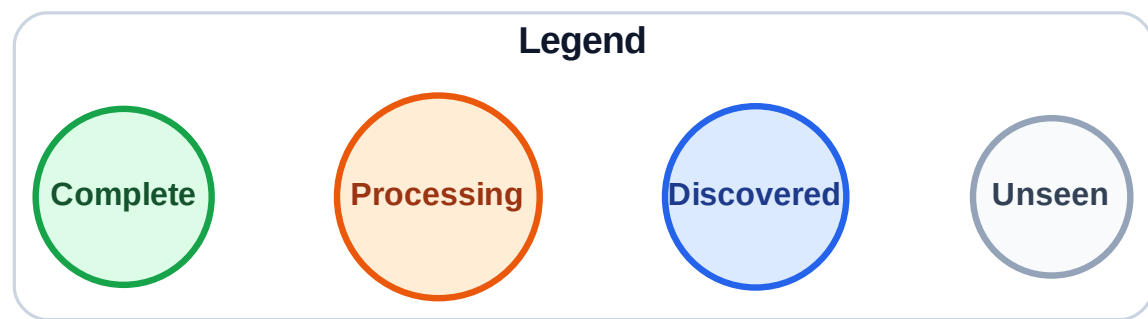
Queue: K → O → D → J → L → F → I

Visited: A, B, C, E



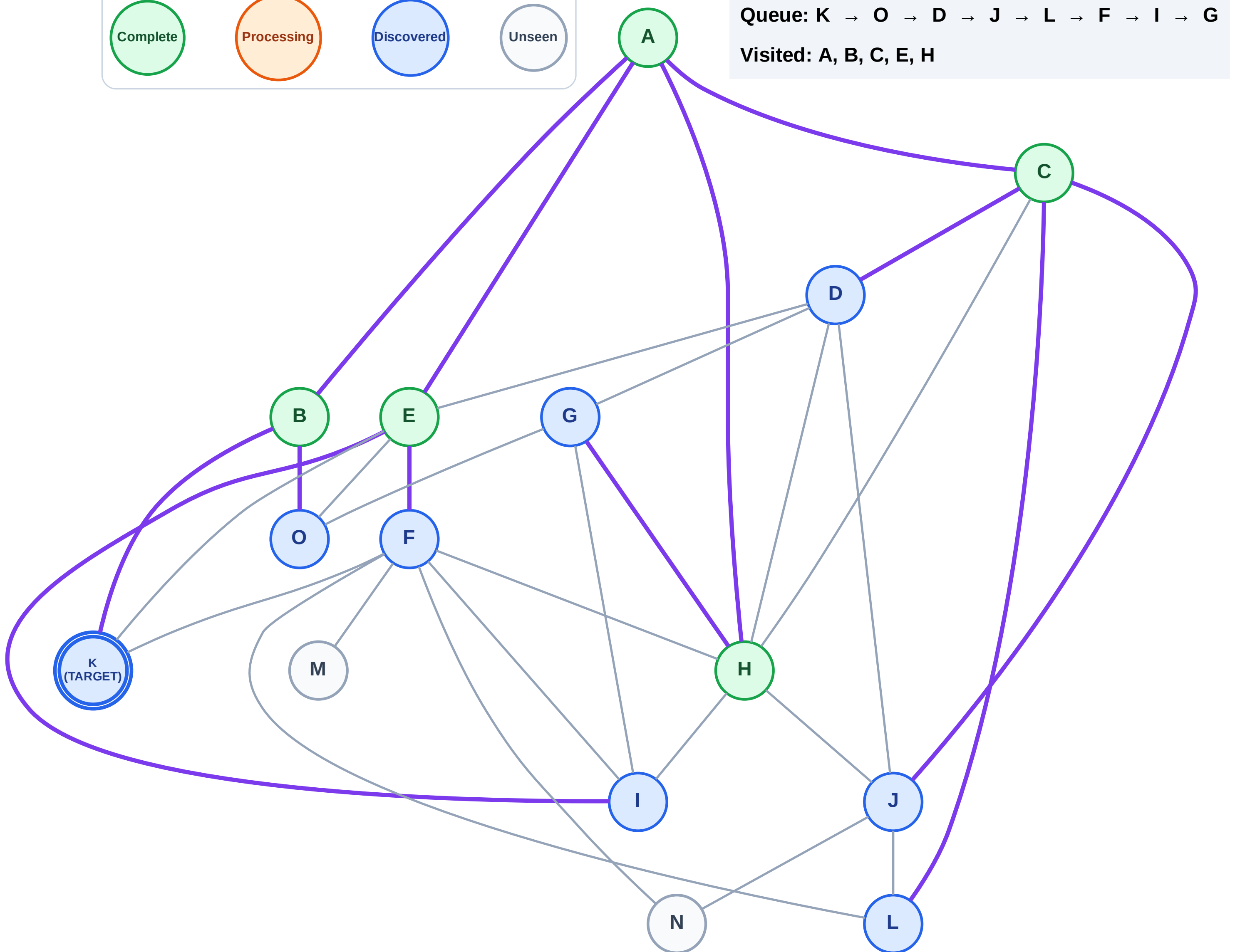
BFS Traversal

Step 11: Discover G from H



Queue: K → O → D → J → L → F → I → G

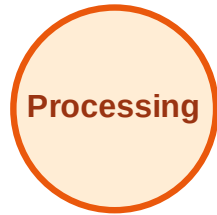
Visited: A, B, C, E, H



BFS Traversal

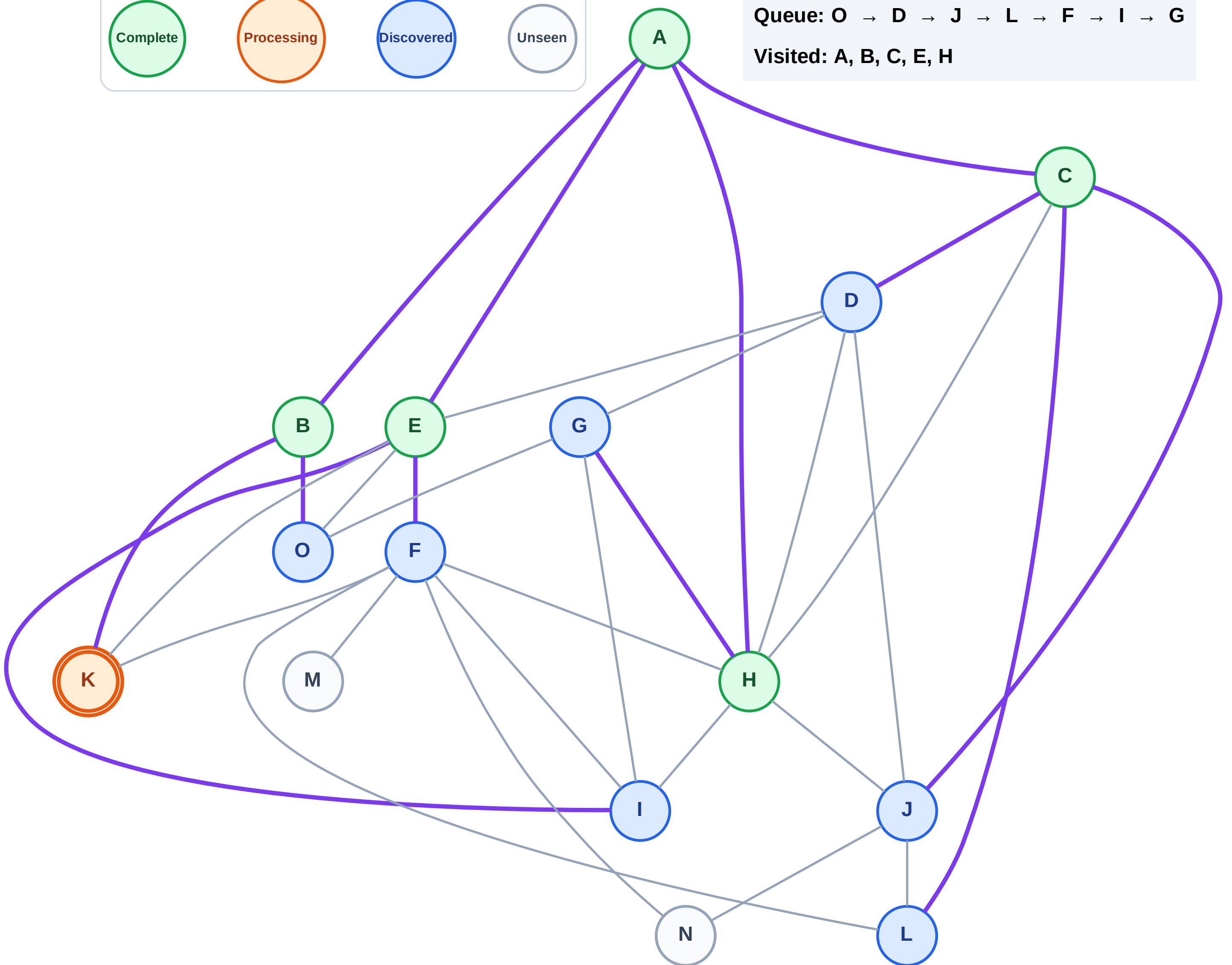
Step 12: Process node K

Legend



Queue: O → D → J → L → F → I → G

Visited: A, B, C, E, H



BFS Traversal

Step 13: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

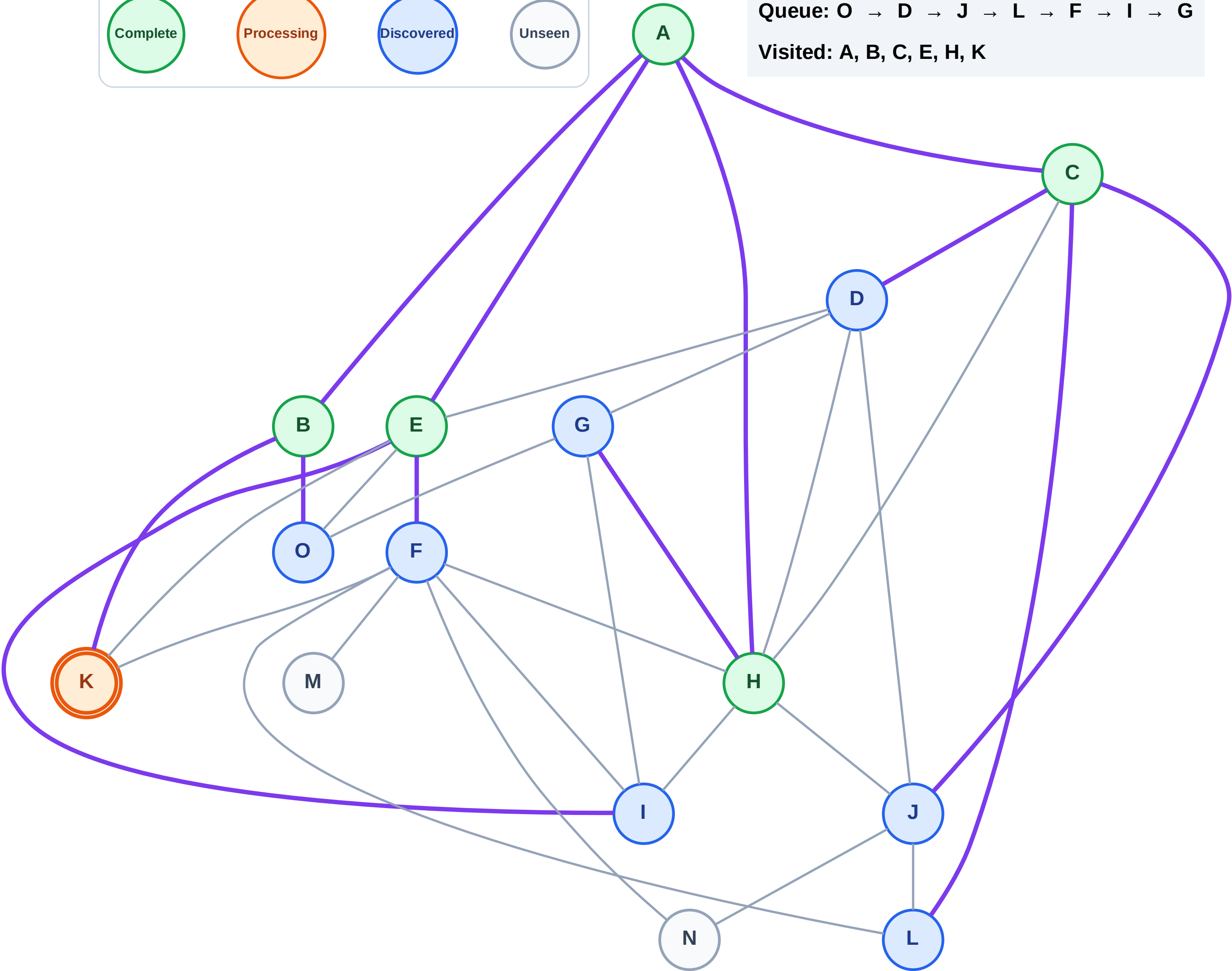
Processing

Discovered

Unseen

Queue: O → D → J → L → F → I → G

Visited: A, B, C, E, H, K



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

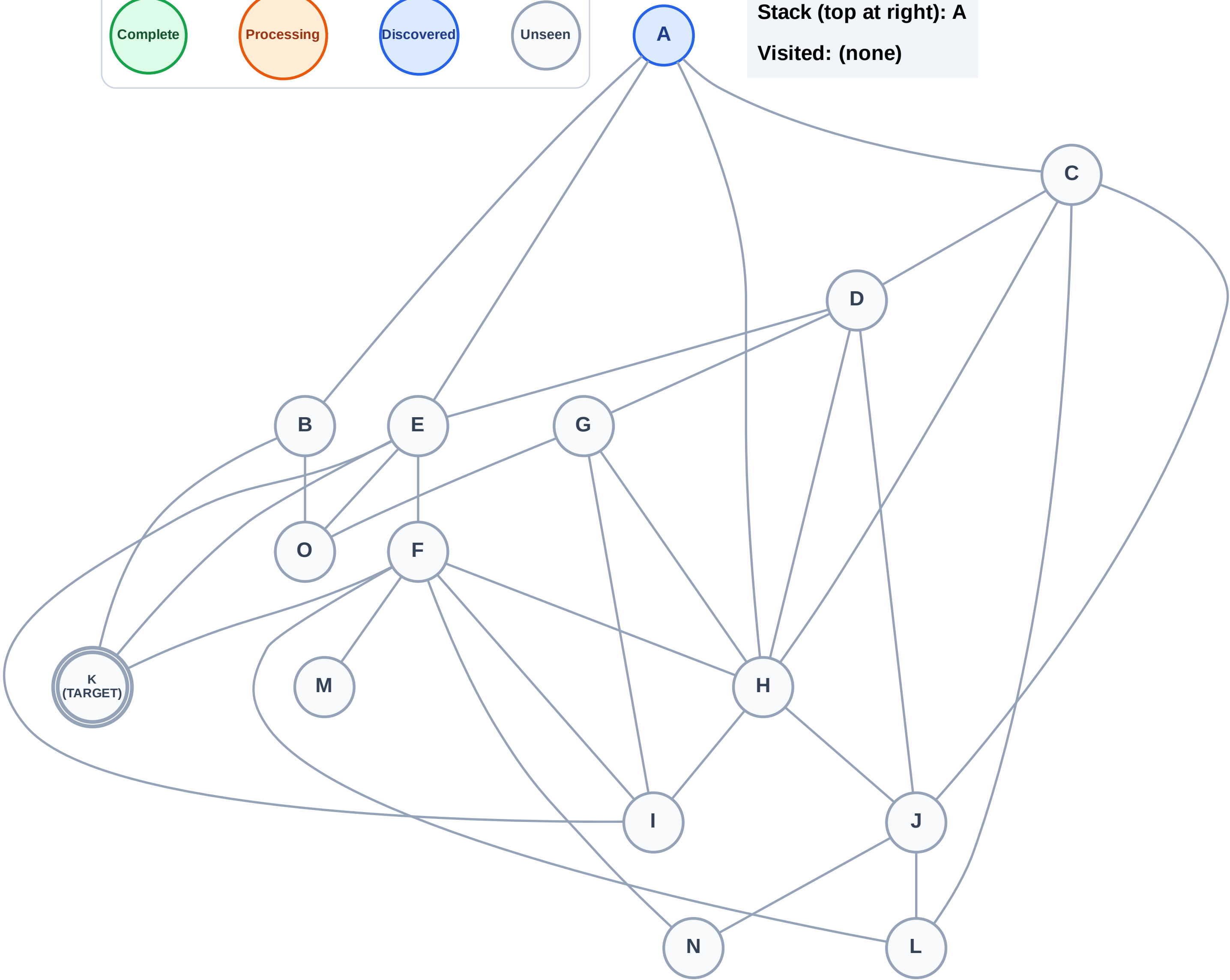
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

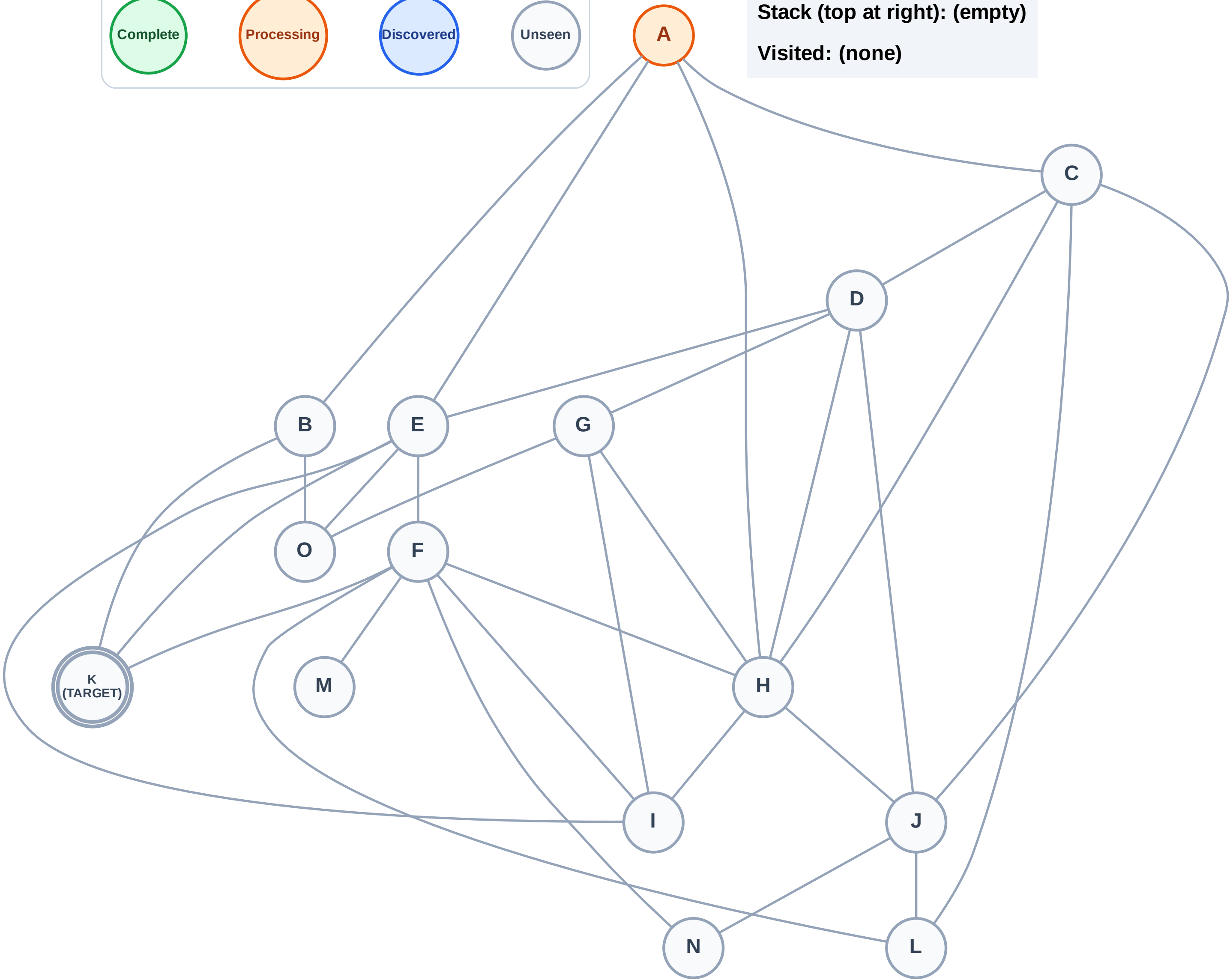
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, E, C, B from A

Legend

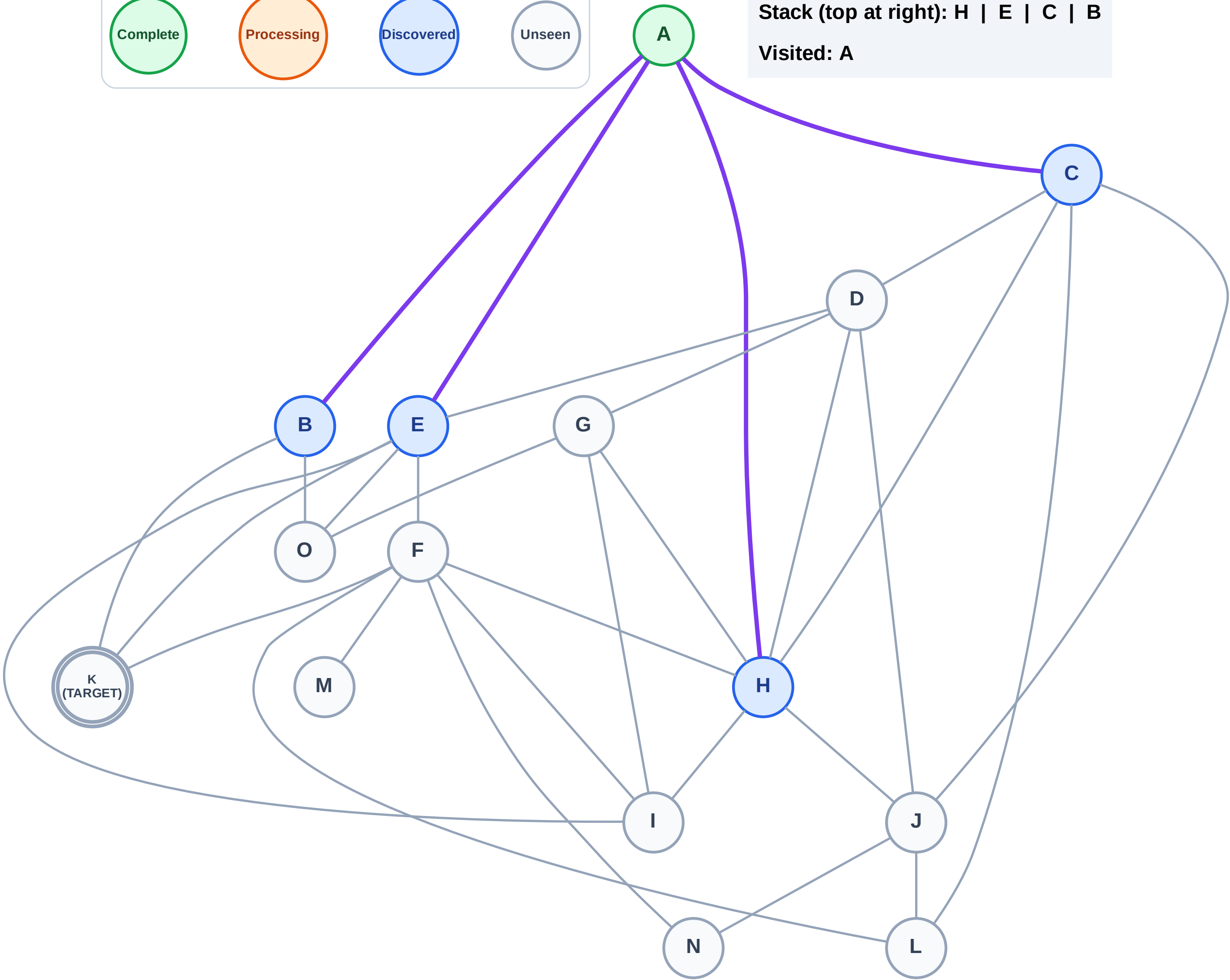
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

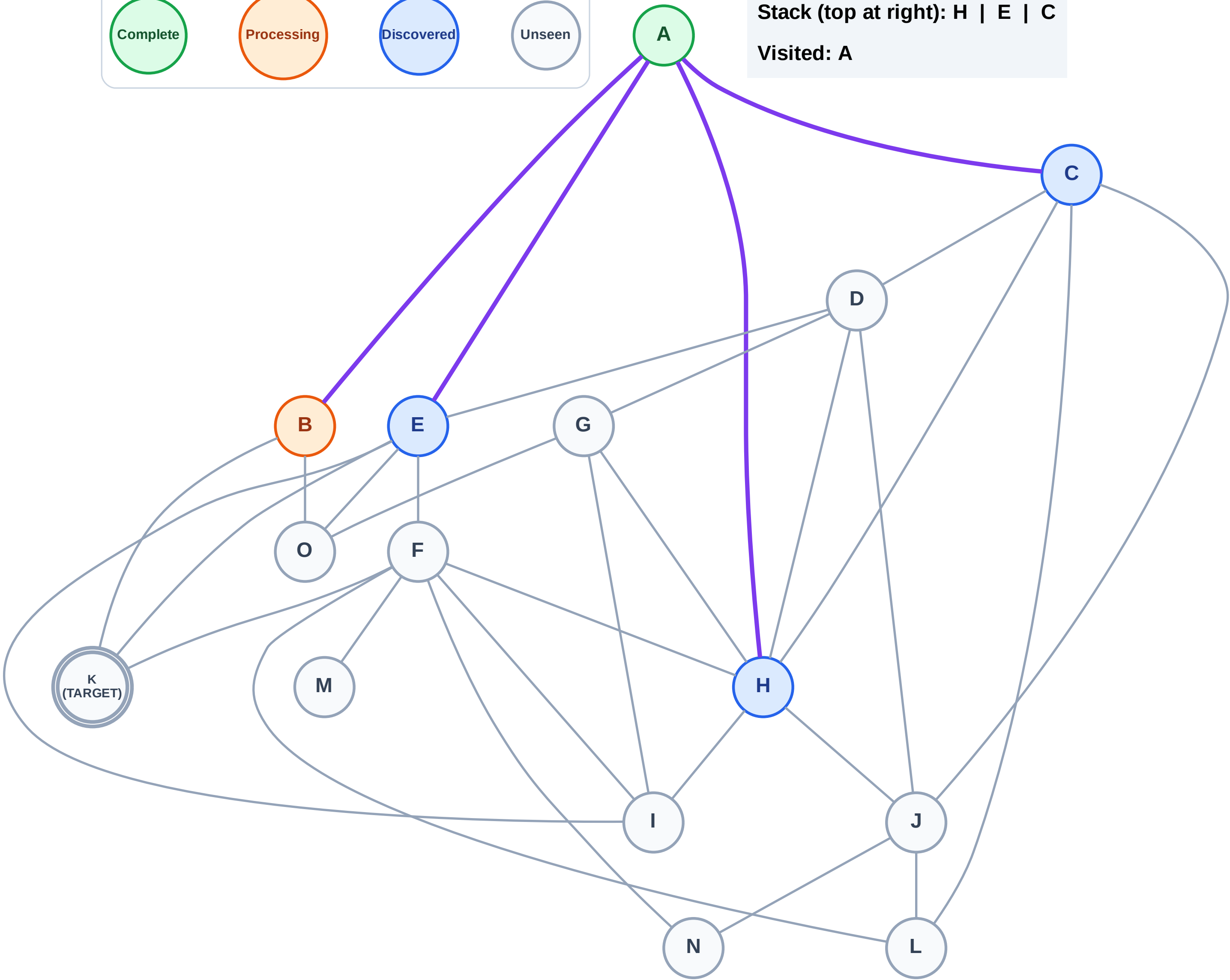
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | C
Visited: A



DFS Traversal

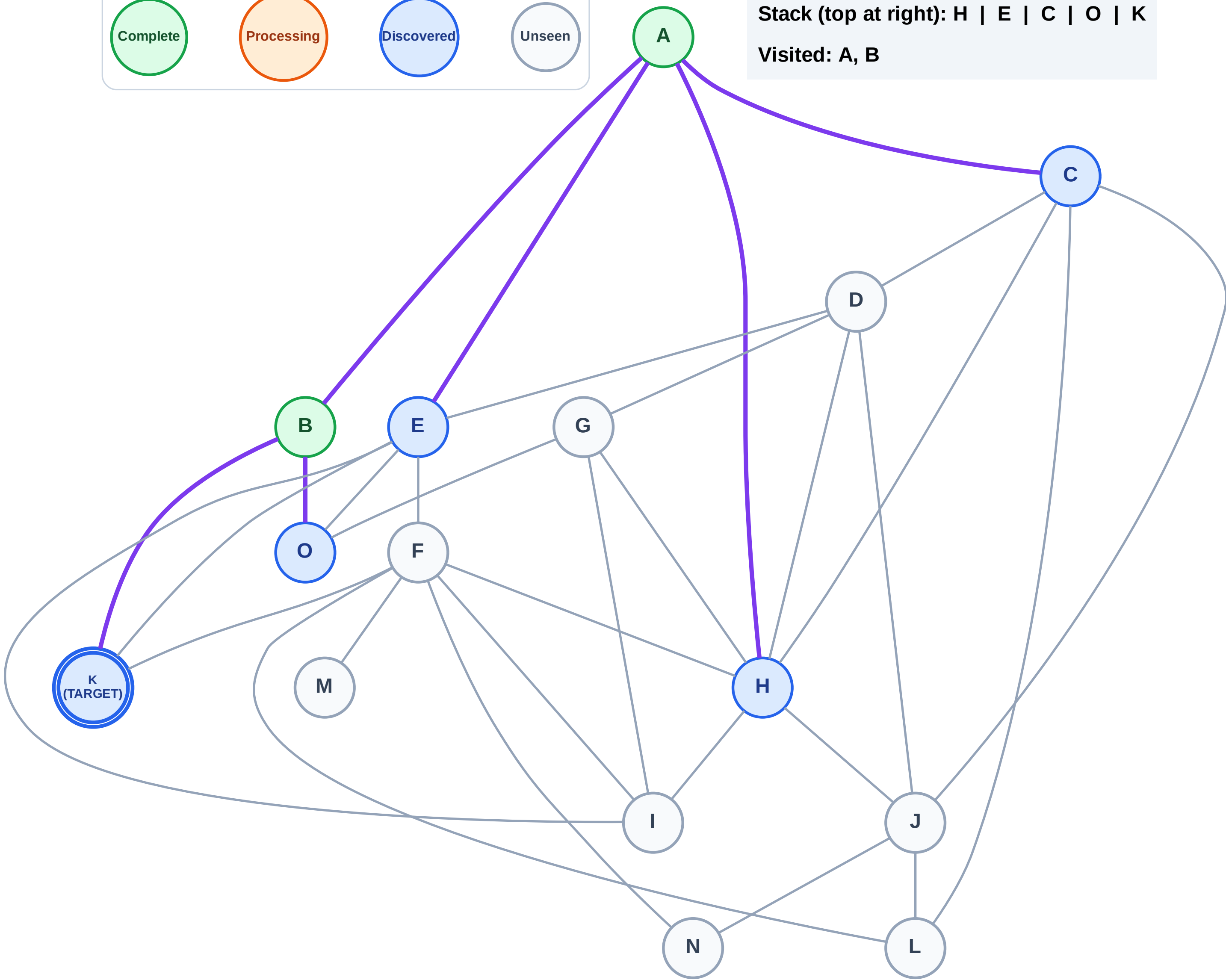
Step 5: Discover O, K from B

Legend



Stack (top at right): H | E | C | O | K

Visited: A, B



DFS Traversal

Step 6: Process node K

Legend

Complete

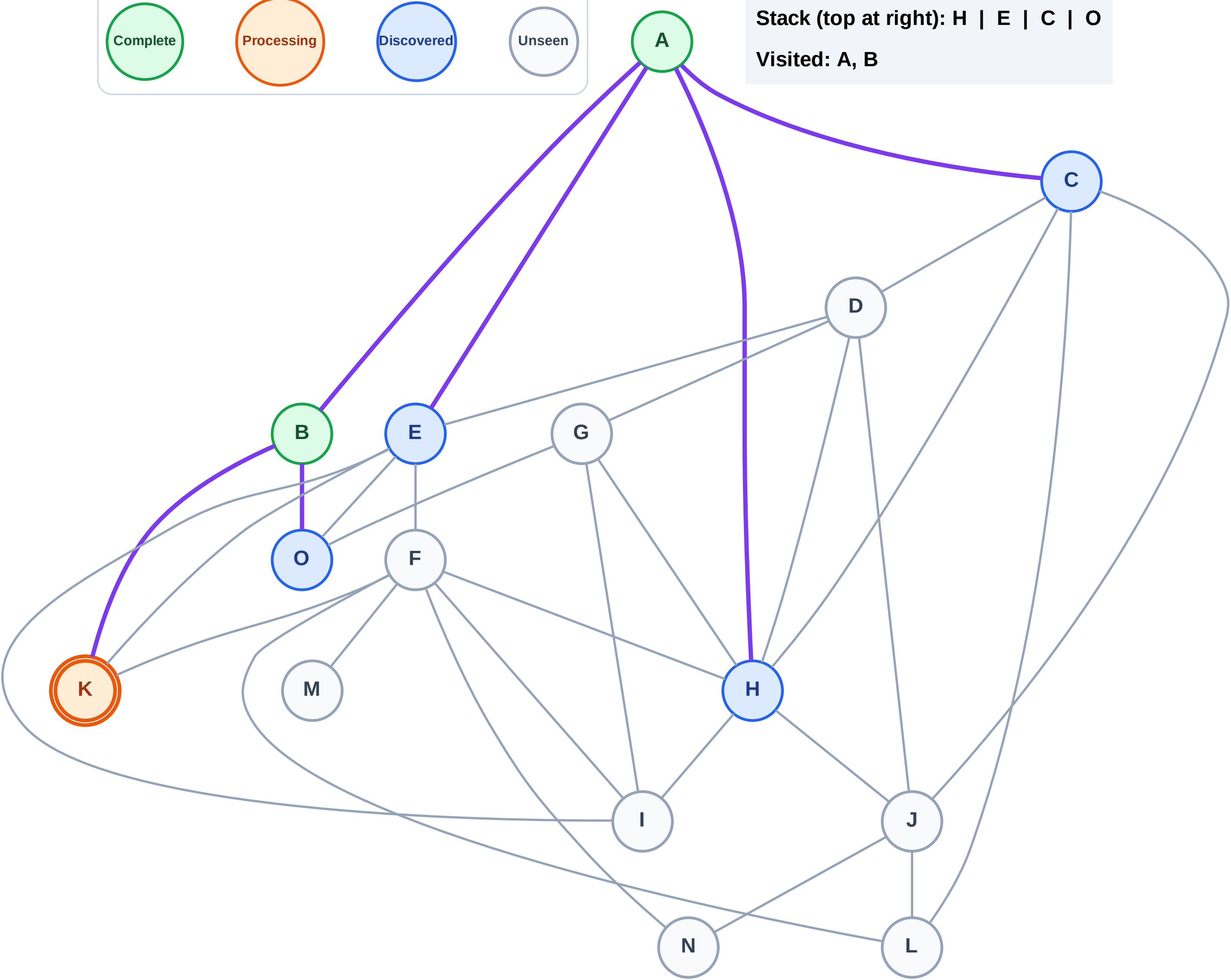
Processing

Discovered

Unseen

Stack (top at right): H | E | C | O

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | C | O
Visited: A, B, K

